Olympiad Test Paper — Science (Class 7)

Chapter 6: Physical and Chemical Changes

Subject	Science (NCERT Class 7)
Chapter	6 — Physical and Chemical Changes
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: **+4** for each correct answer · **−1** for each wrong answer · **0** if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Decide whether a change is physical (no new substance, usually reversible) or chemical (a new substance forms, often shown by heat, light, gas, colour change, or a precipitate). Watch for traps: a change that merely "disappears" or looks dramatic need not be chemical, and one change can show both kinds at once. No calculator is needed. Do not write on this paper — mark answers on your answer sheet.

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1. Which single feature is the **surest** sign that a chemical change has taken place?

(A) the substance changed shape (B) the substance got warmer or cooler
(C) the substance changed from solid to liquid
(D) one or more new substances with new properties were formed

2. A piece of magnesium ribbon is cleaned, then burnt in air with a dazzling white flame, leaving a white powder. The cleaning step and the burning step are, respectively:

(A) physical change and chemical change (B) chemical change and physical change
(C) physical change and physical change (D) chemical change and chemical change

3. Sugar is stirred into a glass of water until it "disappears." This change is:

(A) chemical, because the sugar vanished
(B) chemical, because a new substance (sugar-water) formed
(C) physical, because the sugar can be recovered by evaporating the water
(D) physical, because a gas was given off

4. Which of the following is a **physical** change?

(A) rusting of an iron nail (B) burning of paper
(C) souring of milk (D) melting of candle wax

5. Iron filings are left in moist air for a week and turn into a brown flaky layer. The two things this change needs are:

(A) carbon dioxide and sunlight (B) nitrogen and heat
(C) water only (D) oxygen (from air) and water (moisture)

6. When a burning candle is watched closely, the wax near the flame **melts** and the wick **burns**. These two events are:

(A) both physical changes (B) both chemical changes
(C) a physical change (melting) and a chemical change (burning)
(D) a chemical change (melting) and a physical change (burning)

7. Carbon dioxide gas is bubbled through clear limewater. The limewater:

(A) turns blue (B) turns milky (a white precipitate forms)
(C) turns red (D) stays perfectly clear

8. A clean iron nail is dipped into blue copper sulphate solution. After some time the nail turns brownish and the blue solution fades to pale green. This shows that:

(A) no reaction occurs; the colours are a trick of the light
(B) iron simply dissolves like sugar, a physical change
(C) the nail is being coated with paint
(D) a chemical change occurs — iron displaces copper, forming a new substance

9. Which set lists **only** chemical changes?

(A) melting ice, cutting cloth, dissolving salt
(B) breaking glass, freezing water, bending wire
(C) burning of wood, rusting of iron, cooking of food
(D) tearing paper, boiling water, stretching rubber

10. Crystallisation is best described as a process used to:

(A) burn impurities away with heat
(B) obtain pure crystals of a substance from its solution
(C) turn a solid directly into a gas (D) rust a metal quickly

11. A galvanised bucket lasts far longer in the rain than a plain iron one because galvanising coats the iron with a layer of:

(A) copper (B) gold
(C) zinc (D) plastic paint

12. Which of these everyday processes is a **chemical** change?

(A) ice cubes melting in a drink (B) digestion of food in the stomach
(C) a wet shirt drying on a line (D) chalk being ground into powder

13. Photosynthesis builds sugar from carbon dioxide and water using sunlight. This is classed as a chemical change because:

(A) it is reversible by simply cooling the plant(B) the leaf merely changes colour
(C) no new substance is made

(D) new substances (sugar and oxygen) are formed

14. A student claims "all changes that cannot be reversed are chemical." Which example proves the claim **wrong**?

(A) burning of petrol
(B) cooking of an egg
(C) rusting of a gate
(D) breaking of a glass tumbler

15. Which is the **correct** word equation for rusting?

(A) iron + carbon dioxide → rust
(B) iron + nitrogen → rust
(C) iron + oxygen + water → rust (iron oxide)(D) iron + salt → rust

16. To stop a freshly made iron gate from rusting, the simplest everyday method is to:

(A) paint it
(B) leave it in the rain
(C) sprinkle salt water on it
(D) keep it in moist air

17. Vinegar (an acid) is poured onto baking soda. The fizzing bubbles that form are mainly:

(A) carbon dioxide
(B) oxygen
(C) hydrogen
(D) water vapour

18. A nail kept inside a tightly corked bottle of **dry** air (with a drying agent) does **not** rust, while an identical nail in damp air does. This experiment proves that rusting needs:

(A) only oxygen
(B) both air and moisture
(C) only moisture
(D) sunlight

19. Folding a sheet of aluminium foil into a small cup changes its:

(A) chemical composition
(B) shape only — a physical change
(C) it becomes a new metal
(D) it rusts

20. Burning magnesium gives off so much heat and dazzling **light**. The light and heat are clues that the change is:

(A) chemical
(B) physical
(C) reversible
(D) just melting

21. Which pair correctly matches a change with its type?

(A) curdling of milk — physical
(B) melting of butter — chemical
(C) sublimation of camphor — chemical
(D) ripening of a fruit — chemical

22. In the iron-and-copper-sulphate reaction, the brown coating that appears on the iron nail is:

(A) rust
(B) copper metal
(C) zinc
(D) iron sulphate

23. Which property of rust makes it especially harmful to iron objects?

(A) it is brittle and flaky, so it falls off and exposes fresh iron to keep rusting
(B) it is shiny and hard, protecting the metal(C) it conducts electricity better than iron

(D) it makes the iron stronger

24. Common salt is obtained from seawater mainly by:

(A) burning the water away with fire

(B) letting the water evaporate so salt crystals are left behind

(C) freezing the seawater

(D) bubbling carbon dioxide through it

25. Which observation, on its own, is the **weakest** evidence that a chemical change has occurred?

(A) a gas with a new smell is released

(B) the substance simply changed from solid to liquid

(C) a permanent colour change occurs and cannot be undone

(D) a glowing flame and heat are produced

26. A shopkeeper paints the inside of an iron water tank and keeps the outside oiled. Both painting and oiling prevent rust by:

(A) drying the iron out forever

(B) turning the iron into steel

(C) keeping air and moisture away from the iron surface

(D) adding zinc to the iron

27. Steam from a kettle condenses into water droplets on a cold lid. This change is:

(A) physical, because it is just water changing state and is reversible

(B) chemical, because steam looks different from water

(C) chemical, because heat is lost

(D) physical, because a new gas formed

28. Which one of these is **not** a clue normally used to recognise a chemical change?

(A) a precipitate forms

(B) the object is cut into smaller pieces

(C) a gas is evolved

(D) heat or light is given out

29. Hot, saturated copper sulphate solution is allowed to cool slowly in the lab. Blue crystals appear. This is an example of:

(A) crystallisation

(B) rusting

(C) combustion

(D) displacement

30. Which statement about physical and chemical changes is **true**?

(A) every physical change is reversible and every chemical change is irreversible

(B) only chemical changes can give out heat

(C) a new substance forms in a chemical change but not in a physical change

(D) physical changes always form a precipitate

31. A spoonful of common salt is stirred into water and then the water is fully evaporated. At the end you get back:

(A) a brand-new compound

(B) the same salt — showing the change was physical

(C) a gas (D) rust

32. "Galvanising" protects iron even if the zinc layer gets a small scratch, whereas a scratch in plain paint quickly leads to rust underneath. This hints that galvanising is:

(A) exactly the same as painting

(B) more protective, because the zinc layer keeps shielding the iron

(C) useless once scratched (D) a way to make the iron melt

33. Setting off a sparkler on Diwali involves metal salts burning to give coloured light. This is a chemical change mainly because:

(A) new substances form, releasing light and heat

(B) the sparkler changes shape (C) it can be easily reversed

(D) the metal merely melts

34. Which of the following changes is **reversible**?

(A) burning of a matchstick (B) baking a cake from batter

(C) digestion of a chapati

(D) boiling of water to steam and condensing it back

35. A piece of iron, when galvanised, is dipped in molten zinc. The process of dipping and the protection it later gives are, respectively:

(A) a physical coating process that then blocks air and moisture

(B) a chemical change that makes the iron rust faster

(C) crystallisation followed by evaporation(D) burning followed by melting

36. Four jars each hold an iron nail: (P) in dry air with a drying agent, (Q) in boiled water with oil on top, (R) in tap water open to air, (S) in salty water open to air. Which nail rusts the **most**?

(A) S (B) P

(C) Q (D) R

37. Which everyday change is **physical**, even though it looks dramatic?

(A) a firecracker exploding (B) milk turning into curd

(C) dry ice (solid CO_2) turning straight into a gas

(D) wood burning to ash

38. When green plants respire at night, they use oxygen and release carbon dioxide. Respiration is a chemical change because it:

(A) is just a change of state (B) can be reversed by adding water

(C) breaks down sugar to form new substances and release energy

(D) makes no new substance

39. A student mixes two clear solutions and a solid suddenly settles at the bottom. The solid that forms in such a reaction is called a:

(A) crystal of pure salt
(B) physical mixture
(C) rust
(D) precipitate, a clue to a chemical change

40. Which list places the changes correctly as **physical, physical, chemical**?

(A) burning wax, melting wax, dissolving sugar
(B) melting wax, dissolving sugar, burning wax
(C) rusting iron, melting wax, dissolving sugar
(D) cooking egg, burning paper, melting ice

41. Why is the burning of a candle described as showing **both** a physical and a chemical change?

(A) the wax only melts, nothing burns
(B) the wax only burns, nothing melts
(C) the wax melts (physical) and the wax vapour burns to new substances (chemical)
(D) the flame is an illusion

42. In which case would an iron object rust the **slowest**?

(A) kept in a dry, airtight box with silica-gel packets
(B) left outside during the monsoon
(C) splashed daily with seawater
(D) buried in damp soil

43. The white powder left after magnesium ribbon burns is magnesium oxide. Compared with the shiny metal ribbon, this powder has:

(A) exactly the same properties as magnesium
(B) the property of turning back into ribbon on cooling
(C) no change in chemical identity
(D) new, different properties — proof a new substance formed

44. Which one of these is the best example of a chemical change used **on purpose** for cooking?

(A) baking dough into bread
(B) chopping vegetables
(C) grinding wheat into flour
(D) cooling soup in the fridge

45. Salt is added to seawater pans and the Sun's heat is used. The role of the Sun here is to:

(A) chemically change salt into a new substance
(B) make the salt rust
(C) melt the salt
(D) evaporate the water so the salt crystallises out

46. A magnet is brought near a heap of iron filings and they cling to it; the magnet is taken away and they fall. Magnetising and de-magnetising the iron is:

(A) a chemical change
(B) rusting
(C) a physical change — no new substance forms
(D) crystallisation

47. Three nails are tested. Nail 1 (painted) stays shiny; nail 2 (greased) stays shiny; nail 3 (bare, in damp air) turns brown. The painted and greased nails stay shiny because both coatings:

(A) add oxygen to the iron (B) cut off the iron from air and moisture
(C) chemically convert iron to zinc (D) heat the iron

48. Which change would give out a gas **and** is a chemical change?

(A) boiling water to steam (B) opening a fizzy drink and letting it go flat
(C) baking soda reacting with vinegar (D) ice subliming

49. A nail in test tube X (water that was boiled then sealed under oil) does not rust, while a nail in test tube Y (ordinary tap water, open) does. The single factor that test tube X removed was:

(A) the water (B) the iron
(C) the heat (D) the dissolved air (oxygen)

50. Which statement about crystallisation is **correct**?

(A) it is a chemical change that forms a new substance
(B) it only works by burning the solution
(C) it is a physical change used to get pure crystals from a solution
(D) it always produces a gas

51. A blacksmith heats an iron rod until it is red-hot and hammers it into a curved gate bar, then lets it cool. Ignoring any surface scale, the shaping of the rod is mainly a:

(A) chemical change, because the iron got hot
(B) physical change, because it is still iron in a new shape
(C) crystallisation (D) rusting

52. Copper sulphate solution is blue because of dissolved copper. After an iron nail reacts in it, the solution turns pale green. The green colour is due to the new substance:

(A) copper metal (B) iron sulphate
(C) rust (D) zinc oxide

53. Which is the **odd one out** (the only physical change)?

(A) curdling of milk (B) ripening of a banana
(C) burning of a candle wick (D) condensation of water vapour into dew

54. A "displacement" chemical change is best illustrated by:

(A) iron pushing copper out of copper sulphate solution
(B) wax melting on a warm day (C) ice forming on a pond
(D) sugar dissolving in tea

55. Why do food companies pack snacks in airtight, moisture-free pouches (sometimes with a small "do not eat" sachet)?

(A) to make the food look colourful (B) to add oxygen for freshness

(C) to keep out air and moisture and slow chemical spoiling

(D) to help the food rust

56. Two students argue. Riya: "If a change gives off heat it must be chemical." Sam: "Dissolving washing soda in water gives off heat but no new substance forms." The correct conclusion is:

(A) Riya is right; heat always means chemical(B) both changes are chemical

(C) dissolving always makes a new compound

(D) Sam's example shows some physical changes also release heat, so heat alone is not proof

57. Which sequence is correct for getting pure copper sulphate crystals from an impure sample?

(A) dissolve in hot water → filter → cool slowly so crystals form

(B) burn the sample → collect ash (C) leave it to rust → scrape off

(D) hammer it flat → cut into crystals

58. A laboratory observation list reads: "white precipitate, fizzing gas, sharp new smell, large heat released." How many of these are recognised clues of a chemical change?

(A) one (B) two

(C) three (D) four

59. Which statement comparing rusting and burning is **correct**?

(A) both are chemical changes in which a substance combines with oxygen

(B) both are physical changes (C) burning is chemical but rusting is physical

(D) rusting is chemical but burning is physical

60. A change is found to be (i) accompanied by a new colour, (ii) gives off a gas, and (iii) **cannot** be reversed by any simple physical means. The change is **most likely**:

(A) a physical change (B) only a change of state

(C) a chemical change (D) crystallisation

End of question paper. Maximum marks: 240. Marking: +4 correct, −1 wrong, 0 unattempted.

Solutions — Science (Class 7), Chapter 6: Physical and Chemical Changes

Marking: **+4** correct, **−1** wrong, **0** unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	11	C	21	D	31	B	41	C	51	B
2	A	12	B	22	B	32	B	42	A	52	B
3	C	13	D	23	A	33	A	43	D	53	D
4	D	14	D	24	B	34	D	44	A	54	A
5	D	15	C	25	B	35	A	45	D	55	C
6	C	16	A	26	C	36	A	46	C	56	D
7	B	17	A	27	A	37	C	47	B	57	A
8	D	18	B	28	B	38	C	48	C	58	D
9	C	19	B	29	A	39	D	49	D	59	A
10	B	20	A	30	C	40	B	50	C	60	C

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (D) The one decisive test for a chemical change is that one or more new substances with new properties appear. A shape change, a state change (solid→liquid), or merely warming/cooling can all happen in a purely physical change, so none of those alone is proof.

2. (A) Cleaning the ribbon (rubbing off the dull oxide to reveal shiny metal) just removes a surface layer and changes appearance — physical. Burning it combines magnesium with oxygen to make a brand-new white powder, magnesium oxide — chemical. So physical, then chemical.

3. (C) Sugar "disappearing" is a classic trap: it has only dissolved, not reacted. Taste proves it is still sugar, and evaporating the water brings the sugar back — recoverable, no new substance, hence physical.

4. (D) Melting candle wax only changes its state (solid→liquid) and re-solidifies on cooling — no new substance, physical. Rusting, burning and souring of milk all make new substances (chemical).

5. (D) Rusting needs iron to react with both oxygen from the air and water (moisture). Carbon dioxide, sunlight, nitrogen and "water only" are not what drives rusting.

6. (C) Near the flame the solid wax melts to liquid (state change, physical), while the wick and wax vapour burn to give new gases, soot and heat (chemical). The candle famously shows both kinds of change at once.

7. (B) Carbon dioxide reacts with limewater to form an insoluble white solid (calcium carbonate), so the clear liquid turns milky. This milkiness is the standard test for CO_2. It does not turn blue or red, and it does not stay clear.

8. (D) The blue fading to pale green and the brown coating mean new substances have formed: iron displaces copper from copper sulphate (iron is more reactive). This is a chemical change, not dissolving and not painting.

9. (C) Burning of wood, rusting of iron and cooking of food each make new substances — all chemical. The other sets are full of state/shape changes (melting, freezing, tearing, bending) which are physical.

10. (B) Crystallisation is the process of obtaining pure crystals of a substance from its solution (by slow cooling or evaporation). It does not burn impurities, does not sublime a solid, and certainly does not rust metal.

11. (C) Galvanising means coating iron with a layer of zinc. The zinc takes the brunt of the weather and protects the iron beneath, so a galvanised bucket outlasts a plain iron one.

12. (B) Digestion breaks food down into entirely new substances the body can absorb — chemical. Melting ice, a drying shirt and chalk being ground are all physical (state or size changes).

13. (D) Photosynthesis combines carbon dioxide and water (using light) to make new substances — sugar and oxygen. The formation of new substances is exactly what makes it a chemical change; it is not reversible by cooling and is not a mere colour change.

14. (D) The claim "irreversible ⇒ chemical" is false. Breaking a glass tumbler cannot be undone, yet it is still physical — the pieces are all still glass, no new substance formed. Burning, cooking and rusting are genuinely chemical, so they do not disprove the claim.

15. (C) Rust forms when iron combines with oxygen and water to give hydrated iron oxide: iron + oxygen + water → rust. Carbon dioxide, nitrogen and salt are not the reactants (salt only speeds rusting; it is not consumed into rust).

16. (A) The simplest everyday protection is to paint the gate, sealing the iron from air and moisture. Leaving it in rain, sprinkling salt water, or keeping it in moist air would all *encourage* rusting.

17. (A) Baking soda (a base) reacting with vinegar (an acid) is a neutralisation that fizzes out carbon dioxide gas — the same CO_2 that turns limewater milky. Not oxygen, hydrogen, or plain water vapour.

18. (B) The dry-air nail (moisture removed) does not rust; the damp-air nail does. Air is present in both, so it is the *combination* of air (oxygen) and moisture that is required. Hence rusting needs both air and moisture.

19. (B) Folding foil into a cup only rearranges its shape; it is still aluminium with the same chemical make-up. No new metal, no rusting — purely a shape change, so physical.

20. (A) Giving off lots of heat and dazzling light is a hallmark of a chemical change (burning), not of a physical change or simple melting, and burning magnesium certainly cannot be simply reversed.

21. (D) Ripening of a fruit is a chemical change (new substances, new taste and colour) — correctly matched. Curdling of milk is chemical (not physical), melting butter is physical (not chemical), and sublimation of camphor is physical (not chemical), so those three pairs are mismatched.

22. (B) In the iron + copper sulphate reaction, the more reactive iron pushes out copper, which deposits as a reddish-brown coat of copper metal on the nail. It is not rust, zinc, or iron sulphate (the iron sulphate stays dissolved, colouring the solution green).

23. (A) Rust is brittle and flaky, so it crumbles away instead of forming a protective skin. As it flakes off, fresh iron underneath is exposed to air and moisture and keeps rusting — that is why rust is so destructive.

24. (B) Sea water is led into shallow pans and the Sun lets the water evaporate, leaving common-salt crystals behind. It is not burnt away with fire, not frozen, and CO_2 plays no part.

25. (B) A solid simply melting to a liquid is just a state change — by itself the weakest evidence, since it is usually physical. A new-smelling gas, a permanent unrecoverable colour change, or a flame with heat are all far stronger signs of a chemical change.

26. (C) Both paint and oil work the same way: they form a barrier that keeps air and moisture off the iron surface. They do not dry the iron forever, turn it into steel, or add zinc (that would be galvanising).

27. (A) Steam condensing to water droplets is just water changing state (gas→liquid) and is reversed by heating — physical. No new substance forms; "a new gas formed" and the two "chemical" options are wrong.

28. (B) Cutting an object into smaller pieces is a size change with no new substance — a physical change, so it is *not* a clue of a chemical change. Evolving a gas, forming a precipitate, and giving out heat or light are all genuine chemical-change clues.

29. (A) Cooling a hot saturated solution so that pure crystals separate out is crystallisation. There is no metal reacting (not rusting), no burning (not combustion) and no more-reactive metal pushing out another (not displacement).

30. (C) The defining contrast is that a chemical change forms a new substance while a physical change does not. The other statements are false: not every physical change is reversible (broken glass), physical changes can release heat (dissolving), and they do not always form a precipitate.

31. (B) Dissolving then fully evaporating gives back the very same salt, unchanged. Recovering the original substance proves the dissolving was a physical change — not a new compound, gas, or rust.

32. (B) Because zinc keeps shielding the iron even where the coating is scratched (the zinc is sacrificed first), galvanising is more protective than paint, where a scratch lets rust start underneath. It is not identical to painting, not useless when scratched, and does not melt the iron.

33. (A) A sparkler works because metal salts burn to form new substances, releasing light and heat — the signature of a chemical change. It is not merely a shape change, is not easily reversed, and the metal does more than just melt.

34. (D) Boiling water to steam and then condensing it back recovers the original water — a reversible physical change. Burning a matchstick, baking a cake, and digestion all form new substances and cannot be reversed.

35. (A) Galvanising dips the iron in molten zinc to lay down a zinc *coating* — a physical covering process. The coat then protects by blocking air and moisture from the iron. It does not speed rusting, and it is neither crystallisation nor burning.

36. (A) Salty water open to air rusts iron fastest: it has plenty of oxygen *and* moisture, and dissolved salt speeds the process. Dry air (P) and boiled-water-under-oil (Q) lack moisture or air; tap water open to air (R) rusts, but less than salty water (S).

37. (C) Dry ice subliming (solid CO_2 → gas) is a dramatic-looking but purely physical state change — still CO_2, no new substance. A firecracker, curdling milk, and burning wood are all chemical.

38. (C) Respiration breaks down sugar using oxygen to release energy and form new substances (carbon dioxide and water). Making new substances is what makes it chemical;

it is not a mere state change, is not reversed by adding water, and does form new substances.

39. (D) When two solutions mix and an insoluble solid drops out, that solid is a precipitate — a recognised clue that a chemical reaction has occurred. It is not a grown crystal of pure salt, a physical mixture, or rust.

40. (B) Melting wax (state change) and dissolving sugar (recoverable) are both physical; burning wax forms new substances and is chemical — giving physical, physical, chemical. The other lists put a chemical change (burning, rusting, cooking) in a physical slot.

41. (C) A burning candle shows both: the solid wax first melts (physical), then the wax vapour burns with oxygen to make new gases and soot plus heat and light (chemical). It is not "only melts" or "only burns," and the flame is real.

42. (A) Rusting is slowest where air and moisture are kept away: a dry, airtight box with silica-gel (a drying agent) starves the iron of moisture. Monsoon air, daily seawater, and damp soil all supply abundant moisture and oxygen, speeding rust.

43. (D) Magnesium oxide is a dull white powder with new, different properties from the shiny, bendable magnesium ribbon — proof a new substance has formed. It does not share magnesium's properties, cannot turn back into ribbon on cooling, and is chemically different.

44. (A) Baking dough into bread deliberately uses a chemical change — heat sets off reactions that create new substances (the crust, the rise, the flavour). Chopping, grinding and cooling are all physical.

45. (D) In salt pans the Sun's heat evaporates the water, leaving the dissolved salt to crystallise out — a physical separation. The Sun does not chemically alter the salt, rust it, or melt it.

46. (C) Magnetising and then de-magnetising iron filings only changes a physical property (magnetism); the filings are still iron, no new substance forms — physical. It is not a chemical change, rusting, or crystallisation.

47. (B) Paint and grease both stay shiny because each coating cuts the iron off from air and moisture, the two things rust needs. They do not add oxygen, convert iron to zinc, or heat the iron.

48. (C) Baking soda reacting with vinegar both releases a gas (CO_2) *and* forms new substances — a chemical change with a gas. Boiling water and subliming ice release vapour/gas but are physical; a fizzy drink going flat just lets dissolved gas escape (physical).

49. (D) Boiling the water drives out the dissolved air, and the oil seal stops fresh air dissolving back in; the nail has water but almost no oxygen, so it does not rust. The single factor removed compared with the open tube is the dissolved air (oxygen).

50. (C) Crystallisation is a physical change used to obtain pure crystals from a solution — no new substance is created. It does not form a new compound, does not need burning, and does not produce a gas.

51. (B) Hammering red-hot iron into a curved bar only changes its shape; it is still iron afterwards, so the shaping is a physical change. Getting hot does not make it chemical, and it is neither crystallisation nor rusting.

52. (B) After iron displaces copper, the dissolved product left in solution is iron sulphate, which is pale green — that is the source of the green colour. Copper deposits as solid metal on the nail, rust is not formed here, and there is no zinc involved.

53. (D) Condensation of water vapour into dew is just a state change (gas→liquid), the only physical change in the list. Curdling of milk, ripening of a banana, and burning of a wick are all chemical.

54. (A) Displacement is when a more reactive element pushes out a less reactive one — iron pushing copper out of copper sulphate is the textbook example. Wax melting, ice forming and sugar dissolving are all physical, with no displacement.

55. (C) Airtight, moisture-free packing (often with a silica-gel sachet) keeps out air and moisture, slowing the chemical changes that spoil food (oxidation, going rancid, mould). It is not about colour, adding oxygen, or rusting.

56. (D) Sam is right: dissolving washing soda gives out heat yet forms no new substance, so it is physical. This shows that heat release alone does not prove a chemical change — Riya's rule is too strong. Both changes are not chemical, and dissolving does not always make a new compound.

57. (A) To purify by crystallisation you dissolve the impure sample in hot water, filter out insoluble dirt, then cool slowly so pure crystals form (leaving impurities in solution). Burning, rusting, and hammering do not purify into crystals.

58. (D) A white precipitate, a fizzing gas, a sharp new smell, and a large heat release are *all* recognised clues of a chemical change — that is four of them.

59. (A) Both rusting and burning are chemical changes in which a substance combines with oxygen (iron oxidises slowly to rust; fuels oxidise rapidly while burning). So neither is physical, and the two "one is physical" statements are wrong.

60. (C) A new colour plus a gas evolved plus being unrecoverable by simple physical means together point firmly to new substances having formed — a chemical change. A physical change, a mere change of state, or crystallisation would not show all three.

Solutions complete: 60 / 60.